



Animesh Choudhury

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9, Raghunathpur, Purulia, 723133, West Bengal, India (Home)

● WORK EXPERIENCE

12/08/2018 - 14/10/2018 - PALAMPUR, INDIA

JUNIOR RESEARCH FELLOW CSIR - INSTITUTE OF HIMALAYAN BIORESOURCE TECHNOLOGY

Project Title: "Preparation of Spectral Library of Forest tree species found in Himalayan Region"

Collection of spectral signatures of different Himalayan forest tree species using ASD Fieldspec spectroradiometer and correlate that with different bio-physical parameters. Spectral signatures collected were also used for the classification of different tree species using satellite-based hyperspectral data.

21/10/2018 - 30/03/2019 - DEHRADUN, INDIA

JUNIOR RESEARCH FELLOW INDIAN INSTITUTE OF REMOTE SENSING

Project Title: "EOAM: Monitoring and Assessment of Mountain Eco-System and Services in NW Himalayas: Sub theme-4-Water Resource Status and Availability"

Snow cover area mapping of northwest Himalaya using MODIS, Sentinel, Landsat datasets with the help of Google Earth Engine. Data collection, installation, and supervision of automatic weather station (AWS), snow precipitation gauge (SPG), and snow pack analyzer (SPA)

13/06/2019 - 19/01/2020 - VARANASI, INDIA

JUNIOR RESEARCH FELLOW DST-MAHAMANA CENTRE OF EXCELLENCE IN CLIMATE CHANGE RESEARCH, BANARAS HINDU UNIVERSITY

Observation of horizontal and vertical distribution of different air pollutants over Indo-Gangetic plain using ground and satellite-based observations.

21/01/2020 - 07/02/2022 - MIDNAPORE, INDIA

JUNIOR RESEARCH FELLOW DEPARTMENT OF REMOTE SENSING AND GIS, VIDYASAGAR UNIVERSITY

Project Title: "Sub seasonal to Seasonal Forecast of Agricultural Drought and Estimation of Crop Yield Reduction using Machine Learning Approaches"

Long-term monitoring of droughts using different satellite-based indices and Development of ML Model for prediction of Agricultural Droughts.

15/02/2022 - 30/06/2025 - ROURKELA, INDIA

JUNIOR RESEARCH FELLOW NATIONAL INSTITUTE OF TECHNOLOGY

Project Title: "Land-atmosphere interaction during thunderstorms over major urban areas of India using high-resolution mesoscale modeling framework and climatological data sets"

● EDUCATION & TRAINING

2022 - Current - ROURKELA, INDIA

PHD (EARTH AND ATMOSPHERIC SCIENCES)- NATIONAL INSTITUTE OF TECHNOLOGY

Website: <https://www.nitrkl.ac.in/>

2017 - 2018 - DEHRADUN, INDIA

POST GRADUATE DIPLOMA (RS & GIS)- INDIAN INSTITUTE OF REMOTE SENSING

Website: <https://www.iirs.gov.in/>

2015 - 2017 - INDIA

M.SC (ENVIRONMENTAL SCIENCE)- ASUTOSH COLLEGE, CALCUTTA UNIVERSITY

Website: <https://www.asutoshcollege.in/>

2009 - 2013 - INDIA

B.SC (PHYSICS)- RAGHUNATHPUR COLLEGE, BURDWAN UNIVERSITY

Website: <https://www.raghunathpurcollege.ac.in/>

● **LANGUAGE SKILLS**

Mother tongue(s): **BENGALI**

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C2	C2	C2	C2	C2
HINDI	C2	B2	C2	C2	B2

● **SKILLS**

Remote Sensing (Idrisi, Envi, R, Python, SNAP) | Geographic Information system | Data Science | Data Collection, Data Processing, Data Analysis, Data Visualisation | Python | Machine Learning | Deep Learning | Atmospheric Science | GOOGLE EARTH ENGINE | QGIS | ArcGIS

● **PUBLICATIONS**

[Choudhury A, Yadav AC, Bonafoni S. A Response of Snow Cover to the Climate in the Northwest Himalaya\(NWH\) Using Satellite Products. Remote Sensing. 2021; 13\(4\):655.](#)

[Choudhury, A., Dutta, D., Bera, D., Kundu, A., 2021. Regional variation of drought parameters and long-term trends over India using standardized precipitation evapotranspiration index. J. Environ. Manage. 296, 113056.](#)

[Budakoti, S., Singh, C. & Choudhury, A. Transport of a severe dust storm from Middle East to Indian region and its impact on surrounding environment. Int. J. Environ. Sci. Technol. \(2022\).](#)

[Panda, J., Mukherjee, A., Choudhury, A., Biswas, S. \(2023\). Urban Heat: UHI and Heat Stress Threat to Megacities. In: Chatterjee, U., Shaw, R., Kumar, S., Raj, A.D., Das, S. \(eds\) Climate Crisis: Adaptive Approaches and Sustainability. Sustainable Development Goals Series. Springer, Cham.](#)

[Choudhury, A. Drought trend and its association with land surface temperature \(LST\) over homogeneous drought regions of India \(2001–2019\). Discov Water 4, 51 \(2024\).](#)

[Singh, S., Mukherjee, A., Panda, J. et al. Analysis and Forecasting of Temporal Rainfall Variability Over Hundred Indian Cities Using Deep Learning Approaches. Earth Syst Environ 8, 599–625 \(2024\).](#)

[Mukherjee, A., Panda, J., Choudhury, A., Singh, S., & Bhattacharyya, S. \(2024\). Analyzing urban footprints over four coastal cities of India and the association with rainfall and temperature using deep learning models. Urban Climate, 57, 102123.](#)

[Choudhury, A., Panda, J., & Mukherjee, A. \(2024\). BharatBench: Dataset for data-driven weather forecasting over India.](#)

[Choudhury, A., & Panda, J. \(2025\). Development of a Data-driven weather forecasting system over India with Pangu-Weather architecture and IMDAA reanalysis Data.](#)

- **CONFERENCE PAPERS**

SIMULATION OF AN EXTREME DUST EPISODE USING WRF-CHEM

Links https://cas.iitd.ac.in/iasta2018/pdf/E-Proceedings_IASTA-2018.pdf

ROLE OF EARTH OBSERVATION DATA AND HYDROLOGICAL MODELING IN SUPPORTING UN SDGs IN NORTH WEST HIMALAYA

Links <https://www.int-arch-photogramm-remote-sens-spatial-inf-sci.net/XLIII-B3-2020/853/2020/>

MONITORING OF SURFACE WATER OF EAST KOLKATA WETLANDS (1984-2015) WITH GLOBAL SURFACE WATER DATASET

Links <https://a-a-r-s.org/proceeding/ACRS2020/yi3x7w.pdf>

A STEP TOWARDS BUILDING MEDIUM-RANGE DATA-DRIVEN WEATHER FORECASTING SYSTEM OVER INDIA

Links <https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10984239>

- **POSTER, ONLINE TRAINING, WEBINAR**

Poster presented on " Aerosol vertical distribution over lower Gangetic plain using CALIPSO LIDAR" in the Asian Air Pollution Workshop (AAPW-5) held during 5th-7th November, 2019

Attended a webinar on " Geospatial Techniques in drought characterization and impact assessment " organized by Albedo Foundation on 26th September, 2020

Organizing member, Participant and Speaker of the National Workshop on " Drought Monitoring and Early Warning using Geospatial Techniques " organized by Dept. of Remote Sensing and GIS, Vidyasagar University, West Bengal, India on 24th February, 2021

Attended a National webinar on " Environment and Ecosystem Restoration " organized by Dept. of Physics, K.L.E.F., Vijaywada, Andhra Pradesh, India on 5th June, 2021

Attended 5 days online training program on " Machine Learning (ML) and Deep Learning (DL) for Earth Observation (EO) Data and Application " organized by DA-IITC in Collaboration with SAC, ISRO held during 9th-13th August, 2021

Participated in the online course on "Geospatial Modeling Driven Urban Hazard and Risk Analysis", conducted during 13-02-2023 to 17-02-2023 by IIRS, Dehradun

participated in the training on "Capacity Building - Training School on WRF Modeling System" held online from 3rd to 20th April 2023, jointly conducted by C-DAC and SAMA under the National Supercomputing Mission (NSM)

Completed the "Open Classroom Online Lecture Series in Atmospheric and Climate Sciences" organized by SAMA and Centre for Atmospheric Sciences and Climate Studies, SRM institute of Science and Technology, during 7-24 April 2023

Completed the MOOC Machine Learning in Weather & Climate organised by ECMWF in partnership with IFAB during January to April, 2023

Participated in the short term course on "Deep Learning Application for Smart Cities" held at Department of computer science & Engineering, NIT Rourkela from 11-15 September 2023

Completed short-term course on "Recent Trends in Time Series Forecasting using Deep Learning Models" conducted from December 25-29, 2023 at Department of computer science & Engineering, NIT Rourkela

Participated in the National Symposium Tropment-2024 organised by the Indian Meteorological Society from 10-12 December 2024 at NIT, Rourkela

Poster presented on "Regional Data-driven Weather Forecasting Over India with Pangu-weather Architecture and IMDAA Reanalysis" in the 22nd Annual Meeting of the Asia Oceania Geosciences Society held at Sands Expo & Convention Centre Marina Bay Sands Singapore during the period 27 Jul to 01 Aug, 2025
